

Meta-Regression

This vignette introduces the regression functionality of the `dtametaTMB` package for meta-analysis of diagnostic test accuracy (DTA) studies.

Validation

The subgroup HSROC and Reitsma implementations reproduce the Cochrane Handbook RF and anticcp examples closely, including the baseline parameters (HSROC: accuracy, threshold, shape // Reitsma: logit sensitivity and specificity), between-study variance estimates, and subgroup effects on accuracy and threshold. Small numerical differences may arise from optimizer settings, numerical tolerances, and implementation details, but all benchmark estimates and standard errors were recovered to close agreement.

Dummy-coding vs. cell-means parameterization

For subgroup analyses, we distinguish between two equivalent parameterizations of the same model. In the reference-group parameterization, one subgroup serves as the baseline and the remaining subgroup effects are expressed as deviations from this reference. In the group-specific (cell-means) parameterization, each subgroup has its own parameter directly. The former is convenient for testing subgroup differences, whereas the latter is convenient for reporting subgroup-specific estimates and plotting subgroup-specific HSROC curves. The Reitsma subgroup model is fitted twice with both parameterizations. The subgroup HSROC model is estimated using a reference-group parameterization, and subgroup-specific parameters are recovered by evaluating the fitted linear predictors at subgroup-specific design points via Z_{pred} .

How do we include covariates in the Reitsma model?

For sensitivity in study i , we have the number of diseased individuals testing positive: $y_{Ai} \sim \mathcal{B}(n_{Ai}, \pi_{Ai})$.

Similarly for specificity, we have the number of non-diseased individuals testing negative: $y_{Bi} \sim \mathcal{B}(n_{Bi}, \pi_{Bi})$.

Now we introduce a p -dimensional design-vector z_i including study level covariates. Consequently, at the study level, we have

$$\begin{pmatrix} z_i^\top \mu_{Ai} \\ z_i^\top \mu_{Bi} \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} z_i^\top \mu_A \\ z_i^\top \mu_B \end{pmatrix}, \Sigma \right), \quad \text{with} \quad \Sigma = \begin{pmatrix} \sigma_A^2 & \sigma_{AB} \\ \sigma_{AB} & \sigma_B^2 \end{pmatrix}.$$

Let's assume that z_i includes a single binary covariate representing two subgroups. Then using dummy coding with subgroup 1 being the reference, we have

$$z_i^\top \mu_{Ai} = \begin{cases} \mu_{Ai} & \text{for subgroup 1,} \\ \mu_{Ai} + \nu_{A2} & \text{for subgroup 2,} \end{cases} \quad z_i^\top \mu_{Bi} = \begin{cases} \mu_{Bi} & \text{for subgroup 1,} \\ \mu_{Bi} + \nu_{B2} & \text{for subgroup 2,} \end{cases}$$

$$z_i^\top \mu_A = \begin{cases} \mu_A & \text{for subgroup 1,} \\ \mu_A + \nu_{A2} & \text{for subgroup 2,} \end{cases} \quad z_i^\top \mu_B = \begin{cases} \mu_B & \text{for subgroup 1,} \\ \mu_B + \nu_{B2} & \text{for subgroup 2.} \end{cases}$$

```
data("anticcp")
reitsmasub <- fitReitsmaSubgroup(data=anticcp,
                                TP=TP,
                                FP=FP,
                                FN=FN,
                                TN=TN,
                                study=study,
                                subgroup=generation)

reitsmasub
#>
#> Reitsma Subgroup Model
#> -----
#>
#> Number of studies      : 37
#> Number of subgroups   : 2
#> Model fit              : Converged
#> -2 log likelihood      : 533.37
#> AIC                    : 547.37
#> BIC                    : 558.646
#>
#>
#> Use summary() for parameter estimates.
summary(reitsmasub)
#> $estimates
```

```

#>               Estimate Std_Error
#> mu_A.CCP1      -0.09653883 0.22032058
#> mu_B.CCP1       3.44671920 0.29824368
#> mu_A.CCP2       0.86603855 0.12087681
#> mu_B.CCP2       3.01649066 0.16223753
#> sigma2_A.sens   0.35982866 0.10217649
#> sigma2_B.spec   0.53990927 0.18015721
#> sigma_AB        -0.19684497 0.09835341
#> nu_A.CCP2       0.96257151 0.25134200
#> nu_B.CCP2      -0.43020982 0.33771185
#>
#> $sensspec
#>      type      Orig conflevel  CI_Lower  CI_Upper
#> mu_A.CCP1 sens 0.4758840      0.95 0.3708997 0.5830439
#> mu_B.CCP1 spec 0.9691331      0.95 0.9459445 0.9825578
#> mu_A.CCP2 sens 0.7039207      0.95 0.6522909 0.7508130
#> mu_B.CCP2 spec 0.9533136      0.95 0.9369387 0.9655926
#>
#> $RutterGatsonis_recovered
#>      Lambda      Theta      beta sigma2_alpha sigma2_theta
#> CCP1 4.121047 1.278029 0.2028866 0.4878424 0.3188056
#> CCP2 3.727490 1.950971 0.2028866 0.4878424 0.3188056
#>
#> $subgroups
#> [1] "CCP1" "CCP2"

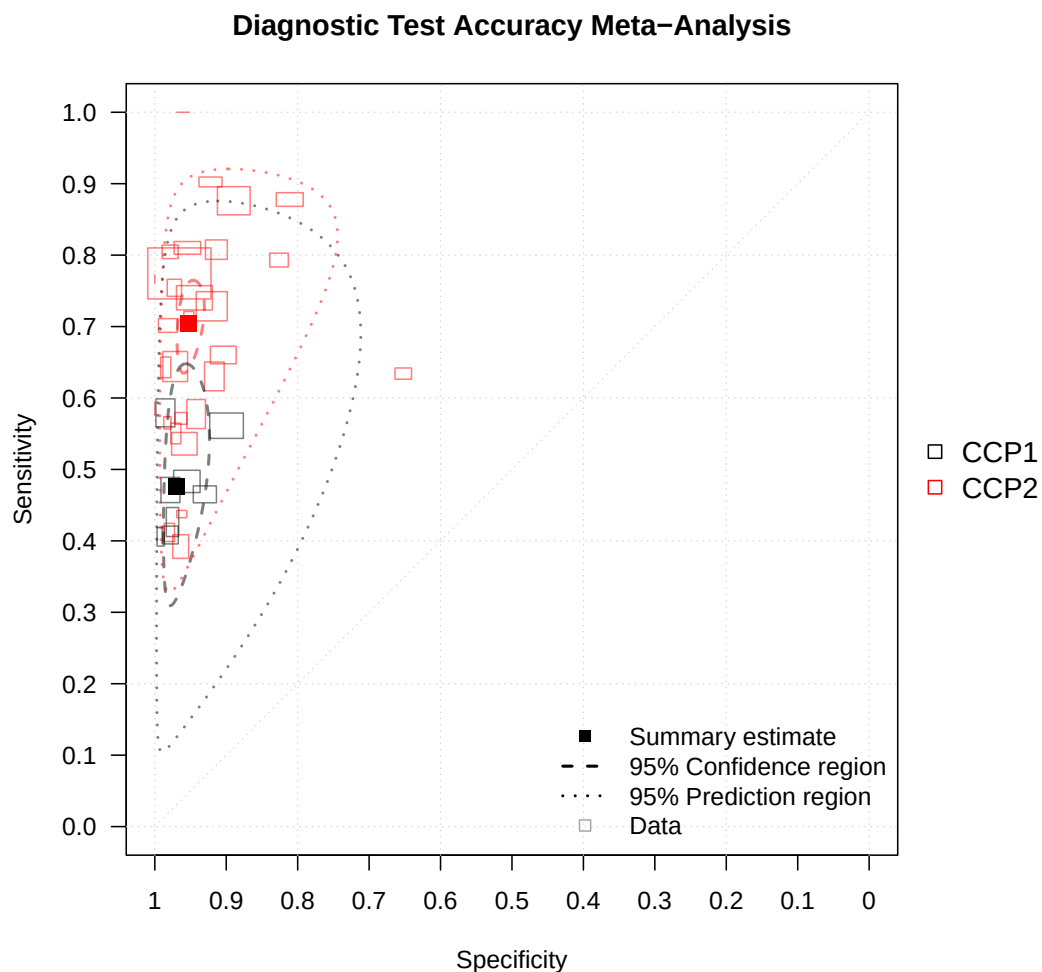
```

How do I get a summary plot of the Reitsma model?

```

plot(reitsmasub,
     scale=0.01,
     nudge_legend=-0.2,
     size="se",
     col=c("black","red"))

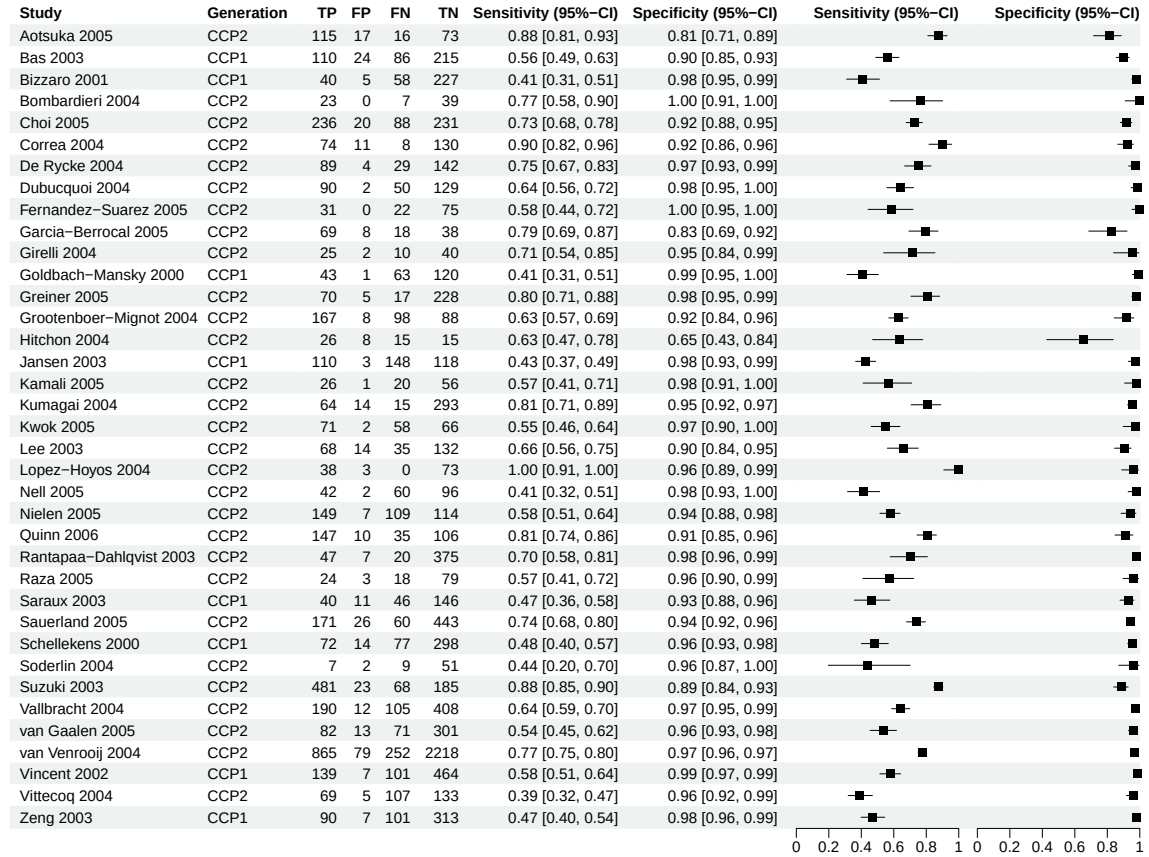
```



How do I get a coupled forest plot?

Note: Rendering forest plots may take longer in an interactive R session due to the underlying grid graphics. Performance is typically faster when knitting the vignette (e.g. via RMarkdown or Quarto).

```
forest(reitsmasub, subgroup_label="Generation")
```



How do I constrain parameters in the Reitsma model?

In sparse data one may wish to fix parameters of the random effects (variance-covariance matrix) at zero. This can be done via the `constrain` argument.

For example,

```
constrainA <- fitReitsmaSubgroup(data=anticcp,
                                TP=TP,
                                FP=FP,
                                FN=FN,
                                TN=TN,
                                study=study,
```

```

                                subgroup=generation,
                                constrain="sigma2_A")
summary(constrainA)$estimates
#>               Estimate Std_Error
#> mu_A.CCP1      -5.439405e-02 0.05498530
#> mu_B.CCP1       3.446625e+00 0.29875482
#> mu_A.CCP2       9.179794e-01 0.03139172
#> mu_B.CCP2       2.996859e+00 0.16216233
#> sigma2_A.sens   4.930381e-32 0.00000000
#> sigma2_B.spec   5.396331e-01 0.17997640
#> sigma_AB        0.000000e+00 0.00000000
#> nu_A.CCP2       9.723735e-01 0.06331527
#> nu_B.CCP2      -4.497661e-01 0.33795382

```

fixes the logit sensitivity variance to zero. This also implies that the random effects covariance is zero. Note that you can also set `constrain` to "sigma2_AB", "sigma2_B", or "all".

Constraining fixed effects is controlled by the `subgroup_constrain` argument. For example,

```

constrainsens <- fitReitsmaSubgroup(data=anticcp,
                                   TP=TP,
                                   FP=FP,
                                   FN=FN,
                                   TN=TN,
                                   study=study,
                                   subgroup=generation,
                                   subgroup_constrain="sens")
summary(constrainsens)$estimates
#>               Estimate Std_Error
#> mu_A.CCP1      0.65330520 0.1274076
#> mu_B.CCP1      3.08122030 0.3256178
#> mu_A.CCP2      0.65330520 0.0000000
#> mu_B.CCP2      3.11800628 0.1745128
#> sigma2_A.sens   0.54192926 0.1462735
#> sigma2_B.spec   0.57765267 0.1996838
#> sigma_AB       -0.27715832 0.1398143
#> nu_A.CCP2      0.00000000 0.0000000
#> nu_B.CCP2      0.03678907 0.3857733

```

assumes equal (logit) sensitivities in all subgroups. Note that you can also set `subgroup_constrain` to "spec" or `c("sens", "spec")`.

How can I compare constrainsens with the full model?

We can perform a likelihood ratio test via `anova`, essentially testing whether there exist subgroup differences in sensitivity.

```
anova(constrainsens,reitsmasub)
#>           Df  logLik Df.diff  Chisq Pr(>Chisq)
#> Model 1    6 -272.77
#> Model 2    7 -266.69      1 12.181  0.0004827 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

How do we include covariates in the Rutter and Gatsonis model?

The number of diseased individuals from study i who test positive is denoted by $y_{i1} \sim \mathcal{B}(n_{i1}, \pi_{i1})$.

Similarly, the number of non-diseased individuals who test positive is $y_{i2} \sim \mathcal{B}(n_{i2}, \pi_{i2})$.

Now we introduce a p -dimensional design-vector z_i including study level covariates. Consequently, at the study level, we have

$$\text{logit}(\pi_{ij}) = (z_i^\top \theta_i + z_i^\top \alpha_i x_{ij}) \exp(-z_i^\top \beta x_{ij}),$$

$$z_i^\top \alpha_i \sim \mathcal{N}(z_i^\top \Lambda, \sigma_\alpha^2), \quad z_i^\top \theta_i \sim \mathcal{N}(z_i^\top \Theta, \sigma_\theta^2),$$

where

$$x_{ij} = \begin{cases} -0.5 & \text{for non-diseased individuals,} \\ 0.5 & \text{for diseased individuals.} \end{cases}$$

Let's assume that z_i includes a single categorical covariate representing three subgroups. Then using dummy coding with subgroup 1 being the reference, we have

$$z_i^\top \theta_i = \begin{cases} \theta_i & \text{for subgroup 1,} \\ \theta_i + \gamma_2 & \text{for subgroup 2,} \\ \theta_i + \gamma_3 & \text{for subgroup 3,} \end{cases} \quad z_i^\top \alpha_i = \begin{cases} \alpha_i & \text{for subgroup 1,} \\ \alpha_i + \xi_2 & \text{for subgroup 2,} \\ \alpha_i + \xi_3 & \text{for subgroup 3,} \end{cases}$$
$$z_i^\top \beta = \begin{cases} \beta & \text{for subgroup 1,} \\ \beta + \delta_2 & \text{for subgroup 2,} \\ \beta + \delta_3 & \text{for subgroup 3,} \end{cases}$$

$$z_i^\top \Lambda = \begin{cases} \Lambda & \text{for subgroup 1,} \\ \Lambda + \xi_2 & \text{for subgroup 2,} \\ \Lambda + \xi_3 & \text{for subgroup 3,} \end{cases} \quad z_i^\top \Theta_i = \begin{cases} \Theta & \text{for subgroup 1,} \\ \Theta + \gamma_2 & \text{for subgroup 2,} \\ \Theta + \gamma_3 & \text{for subgroup 3.} \end{cases}$$

Notes:

- By default, `fitRutterGatsonisSubgroup()` assumes a common shape across all subgroups, which implies $\delta_2 = \delta_3 = 0$ for the case above.
- Moreover, for prediction `fitRutterGatsonisSubgroup()` uses the prediction matrix

$$Z_{\text{pred}} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

for the case above and therefore recovers the threshold, accuracy, and shape parameters for each subgroup as if it were the reference group.

How do I perform a Rutter and Gatsonis meta-regression with a single categorical study-level covariate?

```
data("RF")
RF2      <- RF[RF$method %in% c("LA","ELISA","Nephelometry"),]
RF2$method <- factor(RF2$method,levels=c("LA","ELISA","Nephelometry"))
ruttergatsonissub <- fitRutterGatsonisSubgroup(data=RF2,
                                              TP=TP,
                                              FP=FP,
                                              FN=FN,
                                              TN=TN,
                                              study=study,
                                              subgroup=method,
                                              constrain="shape") # assumes equal shapes in
```

```
ruttergatsonissub
#>
#> Rutter & Gatsonis Subgroup Model
#> -----
#>
#> Number of studies      : 47
#> Number of subgroups   : 3
#> Model fit              : Converged
```



```

#> -2 log likelihood      : 753.722
#> AIC                    : 771.722
#> BIC                    : 788.374
#>
#>
#> Use summary() for parameter estimates.
summary(ruttergatsonissub)
#> $estimates
#>
#>               Estimate Std. Error
#> Lambda_LA      2.45630704 0.32357707
#> xi_ELISA       0.24853054 0.43954126
#> xi_Nephelometry 0.33141751 0.44260069
#> Theta_LA      -0.55005213 0.21334057
#> gamma_ELISA    -0.19625909 0.26096239
#> gamma_Nephelometry 0.49586320 0.26223094
#> beta_LA        0.19768314 0.16983954
#> delta_ELISA     0.00000000 0.00000000
#> delta_Nephelometry 0.00000000 0.00000000
#> sigma2_alpha    1.27791582 0.30852514
#> sigma2_theta    0.47684216 0.11344574
#> Lambda_LA      2.45630704 0.32357707
#> Lambda_ELISA    2.70483758 0.32695434
#> Lambda_Nephelometry 2.78772454 0.30577914
#> Theta_LA      -0.55005213 0.21334057
#> Theta_ELISA    -0.74631122 0.20991330
#> Theta_Nephelometry -0.05418893 0.21213149
#> beta_LA        0.19768314 0.16983954
#> beta_ELISA     0.19768314 0.16983954
#> beta_Nephelometry 0.19768314 0.16983954
#> logitsens      0.82616754 0.28629967
#> logitsens      1.05130793 0.28606921
#> logitsens      1.12639409 0.27689507
#> sens           0.69554397 0.06062755
#> sens           0.74102598 0.05489854
#> sens           0.75517283 0.05119425
#>
#> $sensspec
#>
#>      subgroup      spec conflevel logitsens Std_Error  CI_Lower CI_Upper
#> 1          LA 0.8461538      0.95 0.8261675 0.2862997 0.2650305 1.387305
#> 2          ELISA 0.8461538      0.95 1.0513079 0.2860692 0.4906226 1.611993
#> 3 Nephelometry 0.8461538      0.95 1.1263941 0.2768951 0.5836897 1.669098
#>
#>      Sens SensCI_Lower SensCI_Upper

```

```

#> 1 0.6955440    0.5658725    0.8001616
#> 2 0.7410260    0.6202531    0.8336879
#> 3 0.7551728    0.6419160    0.8414556
#>
#> $Reitsma_recovered
#>          mu_A.sens mu_B.spec sigma2_A.sens sigma2_B.spec  sigma_AB
#> LA          0.6142827 1.962946    0.6534849    0.970378 -0.1573632
#> ELISA        0.5490645 2.316770    0.6534849    0.970378 -0.1573632
#> Nephelometry 1.2135916 1.598491    0.6534849    0.970378 -0.1573632
#>
#> $subgroups
#> [1] "LA"          "ELISA"         "Nephelometry"

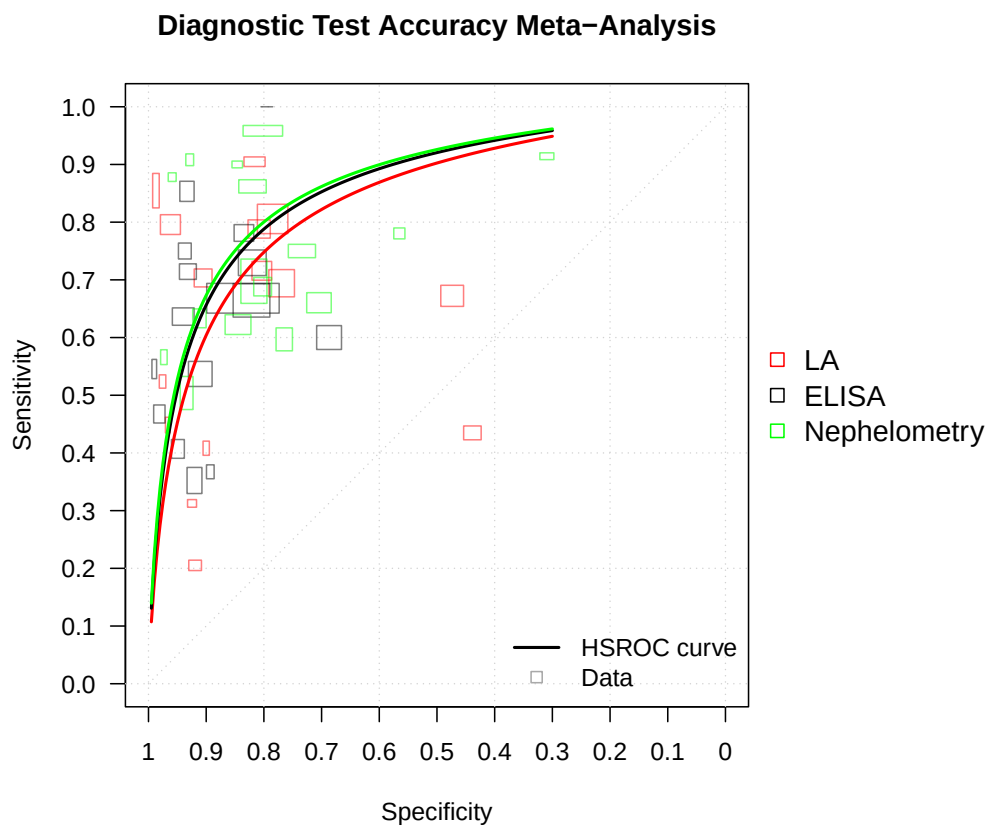
```

How do I get a summary plot of the Rutter and Gatsonis subgroup model?

```

plot(ruttergatsonissub,
     specrange=c(0.3,0.995),
     size="se",
     col=c("red","black","green"),
     scale=0.015)

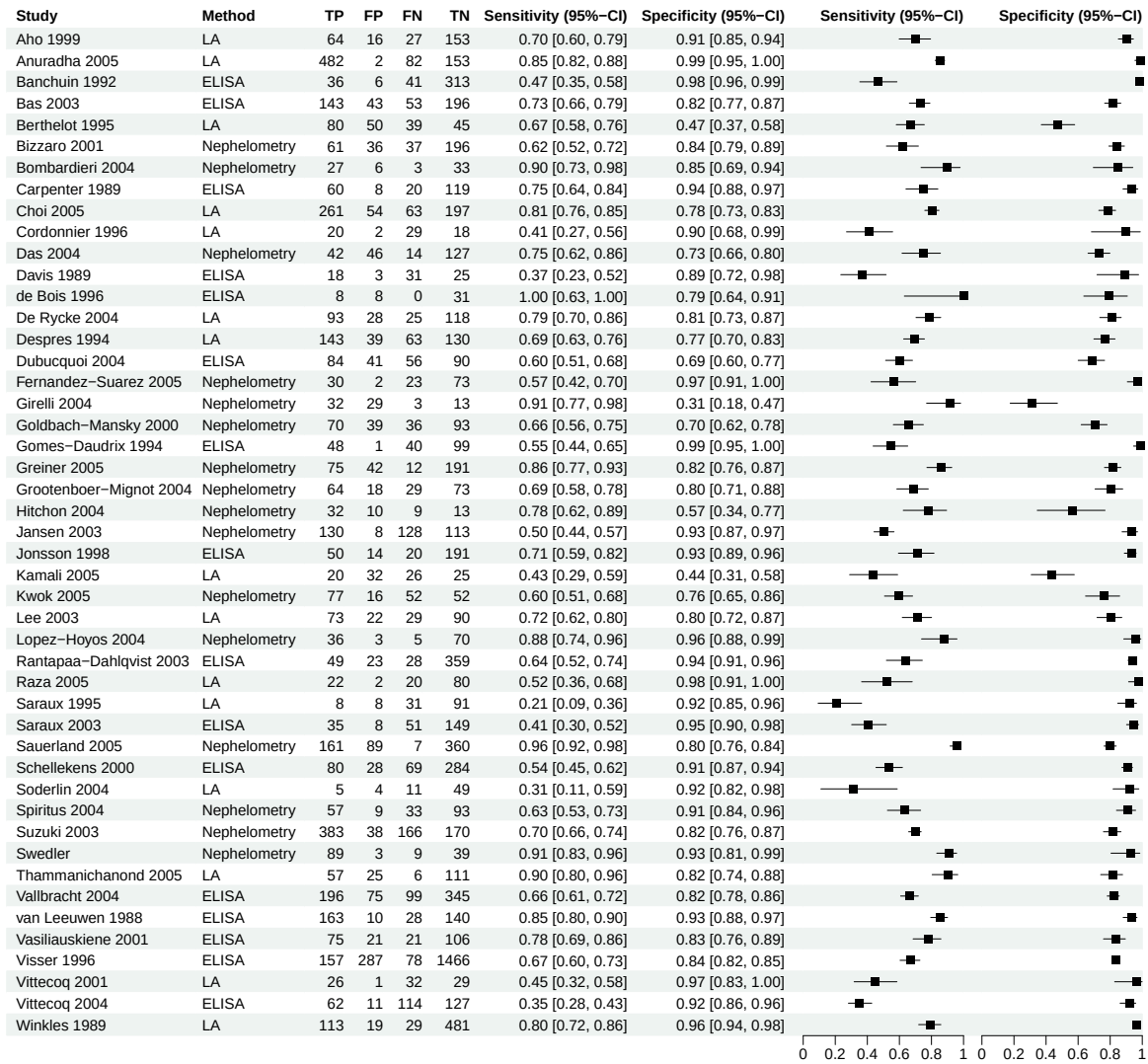
```



How do I get a coupled forest plot?

Note: Rendering forest plots may take longer in an interactive R session due to the underlying grid graphics. Performance is typically faster when knitting the vignette (e.g. via RMarkdown or Quarto).

```
forest(ruttergatsonissub, subgroup_label = "Method")
```



How do I constrain parameters in the Rutter and Gatsonis subgroup model?

In the Rutter and Gatsonis model, all parameter constraints are controlled by `constrain`, i.e.,

- `constrain="sigma2_alpha"` sets σ_{α}^2 to zero,
- `constrain="sigma2_theta"` sets σ_{θ}^2 to zero,

- `constrain="accuracy"` assumes equal accuracy parameters across subgroups, i.e. $\xi_2 = \xi_3 = \dots = 0$,
- `constrain="threshold"` assumes equal threshold parameters across subgroups, i.e. $\gamma_2 = \gamma_3 = \dots = 0$,
- `constrain="shape"` assumes equal shape parameters across subgroups, i.e. $\delta_2 = \delta_3 = \dots = 0$,
- `constrain="shape_zero"` fixes all shape parameters at zero.

Constraints can also be combined, for example `constrain=c("shape", "sigma2_theta")`.

How do I use the general Rutter and Gatsonis regression function?

This method is for advanced users who feel comfortable specifying their own design and prediction matrices. For study-level covariates, the design matrix Z needs two identical consecutive rows per study, one for the diseased and one for the non-diseased. Of note, there are neither `plot()` nor `forest()` methods for `fitRutterGatsonisReg()`.

Let's reproduce the subgroup-analysis from before.

```
# Specify design matrix Z
Z <- model.matrix(~method, data=RF2)
# For study level-covariates, we need to two identical consecutive
# rows per study (diseased and non-diseased).
Z2 <- Z[rep(seq_len(nrow(Z)), each = 2), , drop = FALSE]
# Specify prediction matrix Z_pred
Z_pred <- matrix(c(1,0,0,1,1,0,1,0,1), ncol=3, nrow=3, byrow=T)
constrain <- list(shape_coef=factor(c(1, rep(NA, ncol(Z2) - 1))))

ruttergatsonisreg <- fitRutterGatsonisReg(data=RF2,
                                          TP=TP,
                                          FP=FP,
                                          FN=FN,
                                          TN=TN,
                                          study=study,
                                          Z=Z2,
                                          Z_pred=Z_pred,
                                          map=constrain)

ruttergatsonisreg
#>
#> Rutter & Gatsonis Regression Model
#> -----
#>
```

```

#> Number of studies : 47
#> Model fit          : Converged
#> -2 log likelihood : 753.722
#> AIC                : 771.722
#> BIC                : 788.374
#>
#>
#> Use summary() for parameter estimates.
summary(ruttergatsonisreg)
#> $estimates
#>
#>               Estimate Std. Error
#> accuracy_coef  2.45631156 0.32357673
#> accuracy_coef  0.24852208 0.43954066
#> accuracy_coef  0.33141584 0.44260014
#> threshold_coef -0.55005084 0.21334149
#> threshold_coef -0.19625009 0.26096393
#> threshold_coef  0.49585695 0.26223232
#> shape_coef     0.19768191 0.16983950
#> shape_coef     0.00000000 0.00000000
#> shape_coef     0.00000000 0.00000000
#> sigma2_alpha   1.27791200 0.30852400
#> sigma2_theta   0.47684805 0.11344753
#> Lambda_Pred    2.45631156 0.32357673
#> Lambda_Pred    2.70483365 0.32695391
#> Lambda_Pred    2.78772741 0.30577879
#> Theta_Pred     -0.55005084 0.21334149
#> Theta_Pred     -0.74630093 0.20991415
#> Theta_Pred     -0.05419388 0.21213237
#> beta_Pred      0.19768191 0.16983950
#> beta_Pred      0.19768191 0.16983950
#> beta_Pred      0.19768191 0.16983950
#> logitsens      0.82617129 0.28629952
#> logitsens      1.05130416 0.28606887
#> logitsens      1.12639652 0.27689490
#> sens           0.69554476 0.06062743
#> sens           0.74102525 0.05489857
#> sens           0.75517328 0.05119416
#>
#> $sensspec
#>
#>      spec conflevel logitsens Std_Error  CI_Lower CI_Upper      Sens
#> 1 0.8461538      0.95 0.8261713 0.2862995 0.2650345 1.387308 0.6955448
#> 2 0.8461538      0.95 1.0513042 0.2860689 0.4906195 1.611989 0.7410253

```

```
#> 3 0.8461538      0.95 1.1263965 0.2768949 0.5836925 1.669101 0.7551733
#>   SensCI_Lower SensCI_Upper
#> 1    0.5658735    0.8001621
#> 2    0.6202524    0.8336873
#> 3    0.6419166    0.8414559
```

How do I compare models?

In the previous section we fitted the RF data set using the Rutter and Gatsonis subgroup model while keeping the shape parameter equal across all subgroups `constrain="shape"`. Now let's fit the full model allowing for different shape parameters across subgroups and check whether the data lend support to this approach.

```
ruttermatsonissubfull <- fitRutterGatsonisSubgroup(data=RF2,
                                                    TP=TP,
                                                    FP=FP,
                                                    FN=FN,
                                                    TN=TN,
                                                    study=study,
                                                    subgroup=method,
                                                    constrain=NULL)

ruttermatsonissubfull
#>
#> Rutter & Gatsonis Subgroup Model
#> -----
#>
#> Number of studies      : 47
#> Number of subgroups    : 3
#> Model fit              : Converged
#> -2 log likelihood      : 753.553
#> AIC                   : 775.553
#> BIC                   : 795.905
#>
#>
#> Use summary() for parameter estimates.
summary(ruttermatsonissubfull)
#> $estimates
#>
#>           Estimate Std. Error
#> Lambda_LA      2.4243627 0.33049053
#> xi_ELISA        0.2974489 0.51315108
#> xi_Nephelometry 0.3716072 0.45086464
```

```

#> Theta_LA -0.5029491 0.24451535
#> gamma_ELISA -0.2578309 0.37024437
#> gamma_Nephelometry 0.3970582 0.35944143
#> beta_LA 0.2777423 0.26642216
#> delta_ELISA -0.1028861 0.42660921
#> delta_Nephelometry -0.1603851 0.39753663
#> sigma2_alpha 1.2730509 0.30798445
#> sigma2_theta 0.4762342 0.11343407
#> Lambda_LA 2.4243627 0.33049053
#> Lambda_ELISA 2.7218116 0.39299371
#> Lambda_Nephelometry 2.7959699 0.30708020
#> Theta_LA -0.5029491 0.24451535
#> Theta_ELISA -0.7607799 0.27816983
#> Theta_Nephelometry -0.1058909 0.26322751
#> beta_LA 0.2777423 0.26642216
#> beta_ELISA 0.1748562 0.33321260
#> beta_Nephelometry 0.1173572 0.29500266
#> logitsens 0.8186872 0.27519077
#> logitsens 1.0626859 0.32405595
#> logitsens 1.1206500 0.28834948
#> sens 0.6939576 0.05844514
#> sens 0.7432035 0.06184675
#> sens 0.7541093 0.05346821
#>
#> $sensspec
#> subgroup spec conflevel logitsens Std_Error CI_Lower CI_Upper
#> 1 LA 0.8461538 0.95 0.8186872 0.2751908 0.2793232 1.358051
#> 2 ELISA 0.8461538 0.95 1.0626859 0.3240560 0.4275479 1.697824
#> 3 Nephelometry 0.8461538 0.95 1.1206500 0.2883495 0.5554954 1.685805
#> Sens SensCI_Lower SensCI_Upper
#> 1 0.6939576 0.5693803 0.7954428
#> 2 0.7432035 0.6052880 0.8452503
#> 3 0.7541093 0.6354096 0.8436716
#>
#> $Reitsma_recovered
#> mu_A.sens mu_B.spec sigma2_A.sens sigma2_B.spec sigma_AB
#> LA 0.6172733 1.970644 0.6018251 1.0488519 -0.1579715
#> ELISA 0.5498862 2.315531 0.6670420 0.9463054 -0.1579715
#> Nephelometry 1.2184573 1.594762 0.7065203 0.8934286 -0.1579715
#>
#> $subgroups
#> [1] "LA" "ELISA" "Nephelometry"

```


How do I get the log likelihood, the AIC, and the BIC of a model?

```
logLik(ruttergatsonissubfull)
#> 'log Lik.' -376.7765 (df=11)
AIC(ruttergatsonissubfull)
#> [1] 775.5531
BIC(ruttergatsonissubfull)
#> [1] 795.9047

logLik(ruttergatsonissub)
#> 'log Lik.' -376.8612 (df=9)
AIC(ruttergatsonissub)
#> [1] 771.7223
BIC(ruttergatsonissub)
#> [1] 788.3736
```

How do I perform a likelihood ratio test of the two models?

The test below formally investigates whether the HSROC curve shapes are equal in all sub-groups.

```
anova(ruttergatsonissub,
      ruttergatsonissubfull)
#>           Df  logLik Df.diff  Chisq Pr(>Chisq)
#> Model 1   9 -376.86
#> Model 2  11 -376.78      2 0.1692      0.9189
```

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